

Homework 01 Solutions

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Task 1.1

For an n -dimensional sample space (i.e., $x_0 \in \mathbb{R}^n$), we have that

$$\begin{aligned} W_2^2(\mathcal{N}(0, \mathbf{I}), \mathcal{N}(\sqrt{\bar{\alpha}_T} \mathbf{x}_0, (1 - \bar{\alpha}_T) \mathbf{I})) &= \|\sqrt{\bar{\alpha}_T} \mathbf{x}_0\|^2 + n(1 + (1 - \bar{\alpha}_T) - 2(1 - \bar{\alpha}_T)^{1/2}) \\ &= \bar{\alpha}_T \|\mathbf{x}_0\|^2 + n(2 - \bar{\alpha}_T - 2\sqrt{1 - \bar{\alpha}_T}). \end{aligned}$$

As $\bar{\alpha} \rightarrow 0$, the Wasserstein distance also approaches 0.

Task 1.2

We have

$$\begin{aligned} \log \bar{\alpha}_T &= \sum_{t=1}^T \log(1 - \beta_t) \geq -\sum_{t=1}^T \frac{\beta_t}{1 - \beta_t} \geq -\frac{1}{1 - \beta_{\max}} \sum_{t=1}^T \beta_t = -\frac{\frac{T}{2}(\beta_{\min} + \beta_{\max})}{1 - \beta_{\max}} \\ \therefore \bar{\alpha}_T &\geq \exp\left(-\frac{\frac{T}{2}(\beta_{\min} + \beta_{\max})}{1 - \beta_{\max}}\right) \end{aligned}$$

as the lower bound on $\bar{\alpha}_T$. The corresponding upper bound (using $\log(1 - x) \leq -x$) is

$$\log \bar{\alpha}_T \leq -\frac{T}{2}(\beta_{\min} + \beta_{\max}).$$

Together, this gives

$$-\frac{\frac{T}{2}(\beta_{\min} + \beta_{\max})}{1 - \beta_{\max}} \leq \log \bar{\alpha}_T \leq -\frac{T}{2}(\beta_{\min} + \beta_{\max}),$$

so $\log \bar{\alpha}_T = -\frac{T}{2}(\beta_{\min} + \beta_{\max})(1 + O(\beta_{\max}))$. If $\beta_{\max} \ll 1$, then the lower bound is approximately equal to the upper bound, since $1 - \beta_{\max} \approx 1$. For constant schedule $\beta' = \frac{1}{2}(\beta_{\min} + \beta_{\max})$, we have

$$\log \bar{\alpha}'_T = T \log(1 - \beta') = -T(\beta' + O(\beta'^2)) = -\frac{T}{2}(\beta_{\min} + \beta_{\max})(1 + O(\beta_{\max}^2))$$

This shows that the $\log \bar{\alpha}_T$ decays at the same rate in both schedules to the leading order of $\beta_{\max}$.

Task 1.3

This cosine scheduler is useful because for $1 \leq t \leq T$, $\bar{\alpha}_t$ approach 0 slower compared to a linear scheduler for β_t (Figure 1). This means for fixed T that more steps are spent on denoising meaningful inputs that are not just pure noise. The change in $\bar{\alpha}_t$ is also relatively uniform w.r.t. T , which is necessary to ensure that $p(x_{t-1} | x_t)$ is near-Gaussian. Finally, the rate of change at $t/T \approx 0$ and $t/T \approx 1$ are small, which is useful in the former to ensure fine details in the input are not immediately noised out, and useful in the latter to keep β_t bounded as $\bar{\alpha}_t \rightarrow 0$.

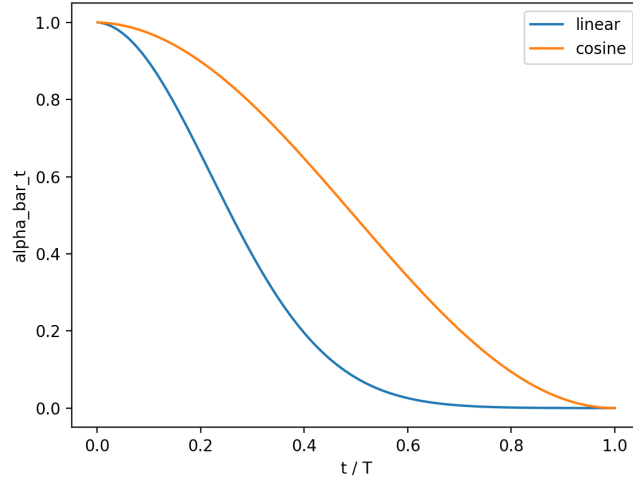


Figure 1: $\bar{\alpha}_t$ vs T for cosine and linear schedules. Parameters: $s = 0.08$, $T = 1000$, $\beta_{\min} = 10^{-4}$, $\beta_{\max} = 0.02$.

Task 2.1

(1) We first show that $\mathbf{P}\{|X_n - X| > \epsilon\} \leq \mathbf{E}[\epsilon^{-1} |X_n - X|]$:

$$\begin{aligned}
 \mathbf{E}[\epsilon^{-1} |X_n - X|] &= \int_{\Omega} \epsilon^{-1} |X_n - X| \, dP(\omega) \\
 &\geq \int_{\Omega} \epsilon^{-1} |X_n - X| I_{|X_n - X| > \epsilon} \, dP(\omega) \\
 &\geq \int_{\Omega} \epsilon^{-1} \cdot \epsilon I_{|X_n - X| > \epsilon} \, dP(\omega) \\
 &= \int_{\Omega} I_{|X_n - X| > \epsilon} \, dP(\omega) \\
 &= \mathbf{P}\{|X_n - X| > \epsilon\}.
 \end{aligned}$$

From Task 2.1.2, we have that $X_n \xrightarrow{L_2} X \Rightarrow X_n \xrightarrow{L_1} X$. For fixed $\epsilon > 0$, we have

$$\begin{aligned} 0 &= \lim_{n \rightarrow \infty} \mathbf{E}[|X_n - X|] = \lim_{n \rightarrow \infty} \mathbf{E}[\epsilon^{-1} |X_n - X|] \\ &= \limsup_{n \rightarrow \infty} \mathbf{E}[\epsilon^{-1} |X_n - X|] \\ &\geq \limsup_{n \rightarrow \infty} \mathbf{P}\{|X_n - X| > \epsilon\}. \end{aligned}$$

Since $\mathbf{P}\{\cdot\} \geq 0$, we have $\liminf_{n \rightarrow \infty} \mathbf{P}\{|X_n - X| > \epsilon\} = 0$, showing that

$$\lim_{n \rightarrow \infty} \mathbf{P}\{|X_n - X| > \epsilon\}$$

exists and is equal to 0 by the Squeeze theorem, giving $X_n \xrightarrow{p} X$, as desired.

- (2) From Holder's inequality with $a = p/q$, $b = \frac{p}{p-q}$ (satisfying $a, b > 1$ since $p > q$ and $1/a + 1/b = 1$ by construction), we get

$$\mathbf{E}[|X_n - X|^q \cdot 1] \leq \mathbf{E}[|X_n - X|^{q \cdot a}]^{1/a} \cdot (1^b)^{1/b} = \mathbf{E}[|X_n - X|^p]^{q/p}.$$

Since $X_n \xrightarrow{L_p} X$, we have that

$$0 = \lim_{n \rightarrow \infty} \mathbf{E}|X_n - X|^p = \lim_{n \rightarrow \infty} \mathbf{E}[|X_n - X|^p]^{q/p}.$$

Together, this shows that

$$\lim_{n \rightarrow \infty} \mathbf{E}|X_n - X|^q \leq \lim_{n \rightarrow \infty} \mathbf{E}[|X_n - X|^p]^{q/p} = 0.$$

Since $|X_n - X| \geq 0$, the limit exists and is equal to 0.

- (3) The implication $X_n \xrightarrow{L_2} X \Rightarrow X_n \xrightarrow{L_1} X$ follows immediately from Task 2.1.2 and the implication $X_n \xrightarrow{p} X \Rightarrow X_n \xrightarrow{d} X$ is given. For the middle implication, we showed in Task 2.1.1 that the stronger statement of $X_n \xrightarrow{L_1} X \Rightarrow X_n \xrightarrow{p} X$ holds. Together, this shows that the implication chain holds as each step.

Task 2.2

Let $I_{[a,b]}$ denote the indicator variable that is 1 on $[a, b]$ and 0 elsewhere.

- (1) Consider $X_n = n^{1/2} I_{[0, 1/n]}$ and $X = 0$ defined on $U(0, 1)$. We have that

$$\begin{aligned} \lim_{n \rightarrow \infty} \mathbf{E} \left| n^{1/2} I_{[0, 1/n]} \right|^2 &= \lim_{n \rightarrow \infty} 1/n \cdot n = 1 \\ \lim_{n \rightarrow \infty} \mathbf{E} \left| n^{1/2} I_{[0, 1/n]} \right|^1 &= \lim_{n \rightarrow \infty} 1/n \cdot n^{1/2} = 0, \end{aligned}$$

showing that $X_n \xrightarrow{L_1} X$ but not $X_n \xrightarrow{L_2} X$.

(2) Consider $X_n = nI_{[0,1/n]}$ and $X = 0$ defined on $U(0,1)$. We have that

$$\lim_{n \rightarrow \infty} \mathbf{E} |nI_{[0,1/n]}|^1 = \lim_{n \rightarrow \infty} 1/n \cdot n = 1,$$

so X_n is not L_1 convergent to X . Then,

$$\lim_{n \rightarrow \infty} \mathbf{P} \{|X_n - X| > \epsilon\} = \lim_{n \rightarrow \infty} \mathbf{P} \{X_n > \epsilon\} = \lim_{n \rightarrow \infty} \mathbf{P} \{U(0,1) \leq 1/n\} = \lim_{n \rightarrow \infty} 1/n = 0,$$

showing that $X_n \xrightarrow{P} X$.

(3) Let $X = U(\{-1,1\})$ and $X_n = -X$ for all $n \geq 1$. Since X and X_n have the same distribution by symmetry, $X_n \xrightarrow{d} X$ trivially. However, $|X_n - X| = 2$ for all n , so X_n does not converge to X in probability.

Task 3.1

We have that

$$\begin{aligned} x_t &= \sqrt{\alpha_t} \left(\sqrt{\bar{\alpha}_{t-1}} x_0 + \sqrt{1 - \bar{\alpha}_{t-1}} \epsilon' \right) + \sqrt{\beta_t} z_t \\ &= \sqrt{\bar{\alpha}_{t-1}} \alpha_t x_0 + \sqrt{(1 - \bar{\alpha}_{t-1}) \alpha_t} \epsilon' + \sqrt{\beta_t} z_t \\ &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{(1 - \bar{\alpha}_{t-1}) \alpha_t} \epsilon + \beta_t \epsilon \\ &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{\alpha_t - \bar{\alpha}_t} \epsilon + \beta_t \epsilon \\ &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon \end{aligned}$$

where $\epsilon \sim \mathcal{N}(0, I)$, showing the induction hypothesis. In particular, step 2 to 3 follows from the additivity of two independent Gaussians. Then,

$$\mathbf{E}[x_t | x_0] = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \mathbf{E}[\epsilon] = \sqrt{\bar{\alpha}_t} x_0,$$

as desired.

Task 3.2

We have that

$$\text{Cov}(x_t | x_0) = \mathbf{E}[(x_t - \mu)(x_t - \mu)^\top | x_0],$$

where $\mu = \mathbf{E}[x_t | x_0] = \sqrt{\bar{\alpha}_t} x_0$. Then, $x_t - \mu = \sqrt{1 - \bar{\alpha}_t} \epsilon$. This gives

$$\mathbf{E}[(x_t - \mu)(x_t - \mu)^\top | x_0] = \mathbf{E}[(\sqrt{1 - \bar{\alpha}_t} \epsilon)(\sqrt{1 - \bar{\alpha}_t} \epsilon)^\top | x_0] = (1 - \bar{\alpha}_t) \mathbf{E}[\epsilon \epsilon^\top] = (1 - \bar{\alpha}_t) \mathbf{I}.$$

Since each Markov step adds independent Gaussian noise, the distribution of x_t is an affine function of x_0 plus a sum of independent Gaussians, so conditioned on x_0 it is Gaussian. As shown above, the distribution has mean $\sqrt{\bar{\alpha}_t} x_0$ and covariance $(1 - \bar{\alpha}_t) \mathbf{I}$, giving

$$q(x_t | x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t) \mathbf{I}),$$

as desired.

Task 3.3

We have

$$\begin{aligned}
 \mathbf{E}\|x_t\|^2 &= \mathbf{E} \left[(\sqrt{\bar{\alpha}_t}x_0)^\top (\sqrt{\bar{\alpha}_t}x_0) + 2(\sqrt{\bar{\alpha}_t}x_0)^\top (\sqrt{1-\bar{\alpha}_t}\epsilon) + (\sqrt{1-\bar{\alpha}_t}\epsilon)^\top (\sqrt{1-\bar{\alpha}_t}\epsilon) \right] \\
 &= \bar{\alpha}_t \mathbf{E} [x_0^\top x_0] + (1-\bar{\alpha}_t) \mathbf{E} [\epsilon^\top \epsilon] \\
 &= \bar{\alpha}_t d + (1-\bar{\alpha}_t)d \\
 &= d.
 \end{aligned}$$

The removal of the cross term is due to the independence of x_0 and ϵ . As $\bar{\alpha}_T \rightarrow 0$, $q(x_T | x_0)$ approaches $\mathcal{N}(0, \mathbf{I})$. For the given formula for β_t , $\bar{\alpha}_{100} \approx 0.602$, $\sqrt{\bar{\alpha}_{100}} \approx 0.776$, $\sqrt{1-\bar{\alpha}_{100}} \approx 0.630$.

Task 4.1

We have

$$\text{Var}(W_t - W_s) = \text{Cov}(W_t - W_s, W_t - W_s) = \text{Var}(W_t) + \text{Var}(W_s) - 2\text{Cov}(W_s, W_t) = t + s - 2s = t - s.$$

This is consistent with the distribution of $W_t - W_s$ being $\mathcal{N}(0, t - s)$.

Task 4.2

We have

$$\begin{aligned}
 \text{Var}(W_1 + W_2) &= \text{Var}(2W_1 + (W_2 - W_1)) = 4\text{Var}(W_1) + \text{Var}(W_2 - W_1) = 5 \\
 \text{Cov}(W_1, W_3 - W_2) &= 0.
 \end{aligned}$$

The intervals $[0, 1]$ and $[2, 3]$ are disjoint, so the Wiener process increments on those intervals are independent, implying a covariance of 0.

Task 4.3

We have

$$\text{Corr}(W_s, W_t) = \frac{\text{Cov}(W_s, W_t)}{\sqrt{\text{Var}(s)\text{Var}(t)}} = \frac{s}{\sqrt{st}} = \sqrt{s/t}.$$

For $s = 1, t = 4$, $\text{Corr}(W_s, W_t) = 1/2$. As $t \rightarrow \infty$ with s fixed, $\text{Corr}(W_s, W_t) \rightarrow 0$. This shows that earlier values for a Wiener process matter increasingly less as the time window grows, since $\text{Var}(W_t)$ grows without bound and W_s remains constant.